JAVASCRIPT ASSIASSIGNM

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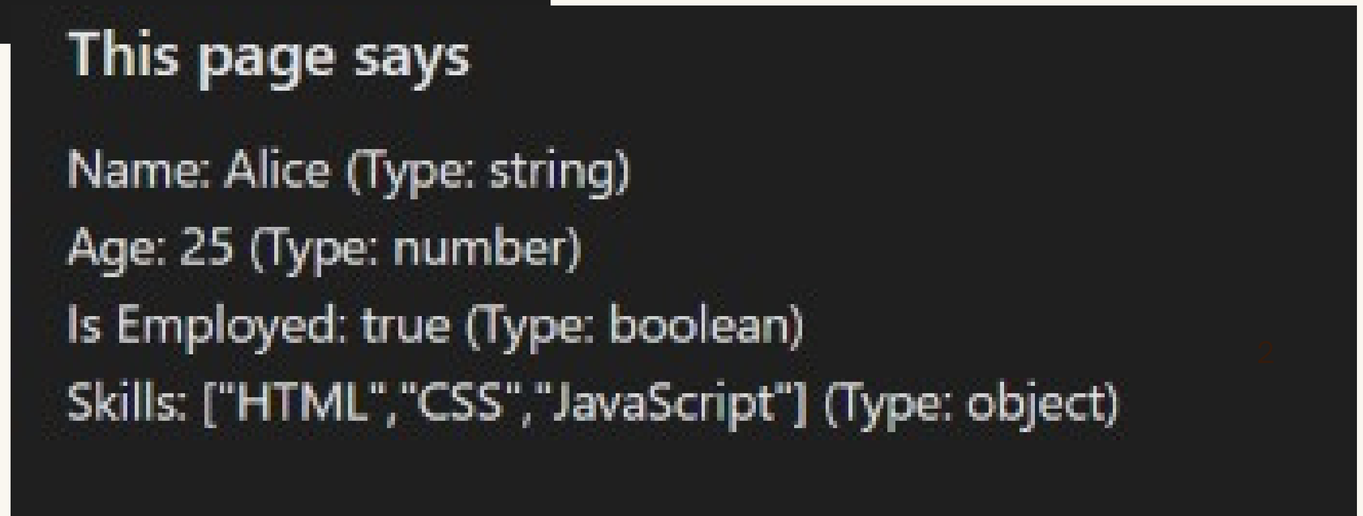
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Assignment 1:vairables,Data Types, and console Operations

Code

```
1   var name = "Alice";
2   let age = 25;
3   const isEmployed = true;
4   const skills = ["HTML", "CSS", "JavaScript"];
5   let output = "Name: " + name + " (Type: " + typeof name + ")\n";
6   output += "Age: " + age + " (Type: " + typeof age + ")\n";
7   output += "Is Employed: " + isEmployed + " (Type: " + typeof isEmployed + ")\n";
8   output += "Skills: " + JSON.stringify(skills) + " (Type: " + typeof skills + ")";
9   console.log(output);
10  alert(output);
```

output



Assignment 2 : Operators,Expressions,and input Handling

Code

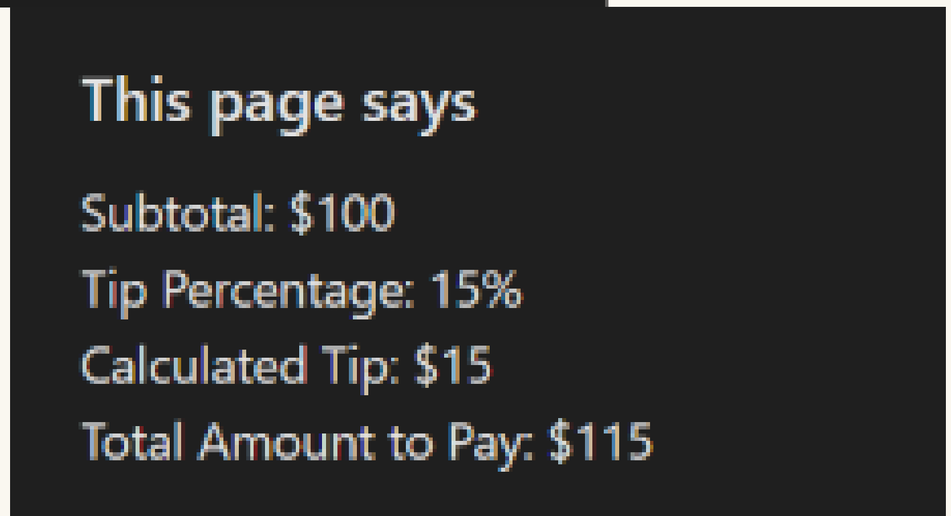
```
let billAmount = Number(prompt("Enter your bill amount:", "100"));
let tipPercentage = Number(prompt("Enter tip percentage:", "15"));

let tipAmount = billAmount * tipPercentage / 100;
let totalAmount = billAmount + tipAmount;

let output =
    "Subtotal: $" + billAmount +
    "\nTip Percentage: " + tipPercentage + "%" +
    "\nCalculated Tip: $" + tipAmount +
    "\nTotal Amount to Pay: $" + totalAmount;

console.log(output);
alert(output);
```

output



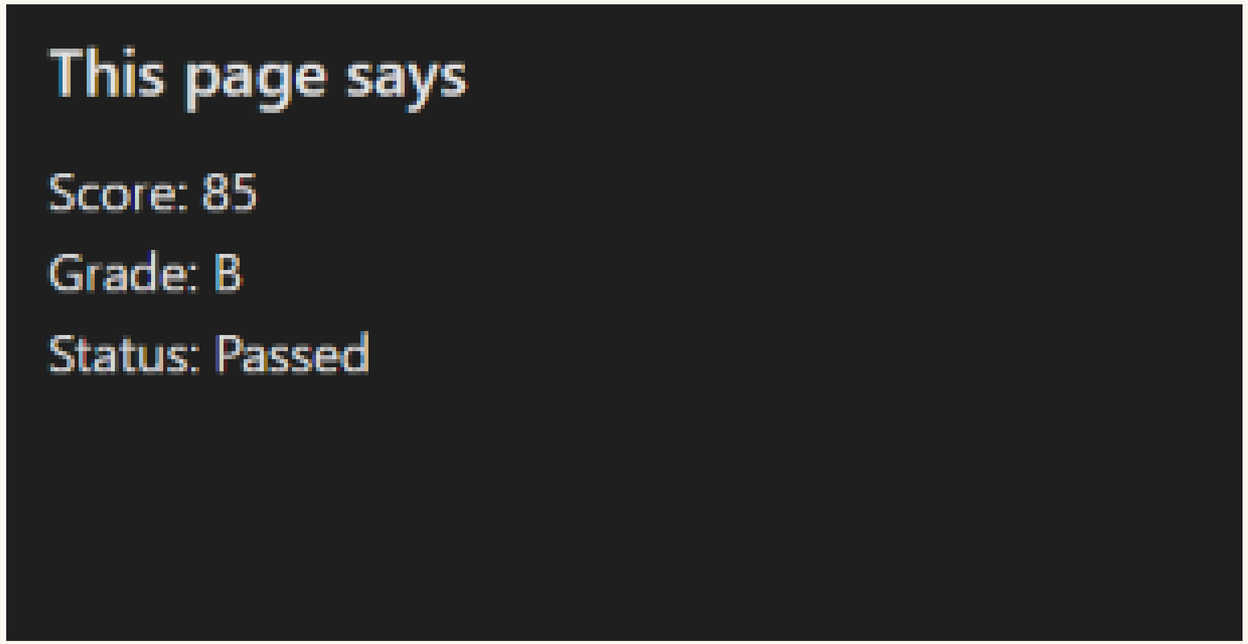
Assignment 3 : Conditional Logic Implementation

Code

```
let score = 85;
let grade;
if (score >= 90) {
    grade = "A";
} else if (score >= 80) {
    grade = "B";
} else if (score >= 70) {
    grade = "C";
} else if (score >= 60) {
    grade = "D";
} else {
    grade = "F";
}
switch (grade) {
    case "A":
        grade = "A";
        break;
    case "B":
        grade = "B";
        break;
    case "C":
        grade = "C";
        break;
    case "D":
        grade = "D";
        break;
    default:
        grade = "F";
}
let status = score >= 40 ? "Passed" : "Failed";
console.log("Score: " + score);
console.log("Grade: " + grade);
console.log("Status: " + status);alert("Score: " + score + "\nGrade: " + grade + "\nStatus: "
+ status);

```

output



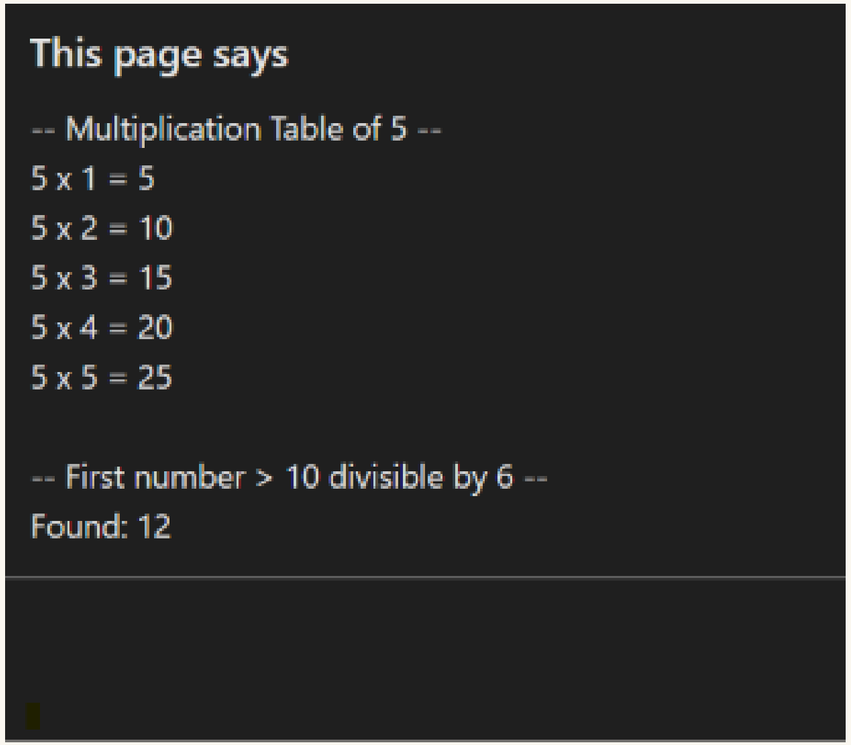
Assignment 4 : Loop control structures

Code

```
console.log("-- Multiplication Table of 5 --");
for (let i = 1; i <= 5; i++) {
    console.log("5 x " + i + " = " + (5 * i));
}
let num = 11;
while (true) {
    if (num > 10 && num % 6 === 0) {
        console.log("-- First number > 10 divisible by 6 --");
        console.log("Found: " + num);
        break;
    }
    num++;
}
console.log("-- Do...While Loop with Continue --");
let j = 0;
do {
    j++;

    if (j === 3) {
        continue;
    }
    console.log("Number: " + j);
} while (j < 5);
alert(
    "Multiplication Table of 5:\n" +
    "5 x 1 = 5\n" +
    "5 x 2 = 10\n" +
    "5 x 3 = 15\n" +
    "5 x 4 = 20\n" +
    "5 x 5 = 25\n\n" +
    "First number > 10 divisible by 6: " + num
);
```

output

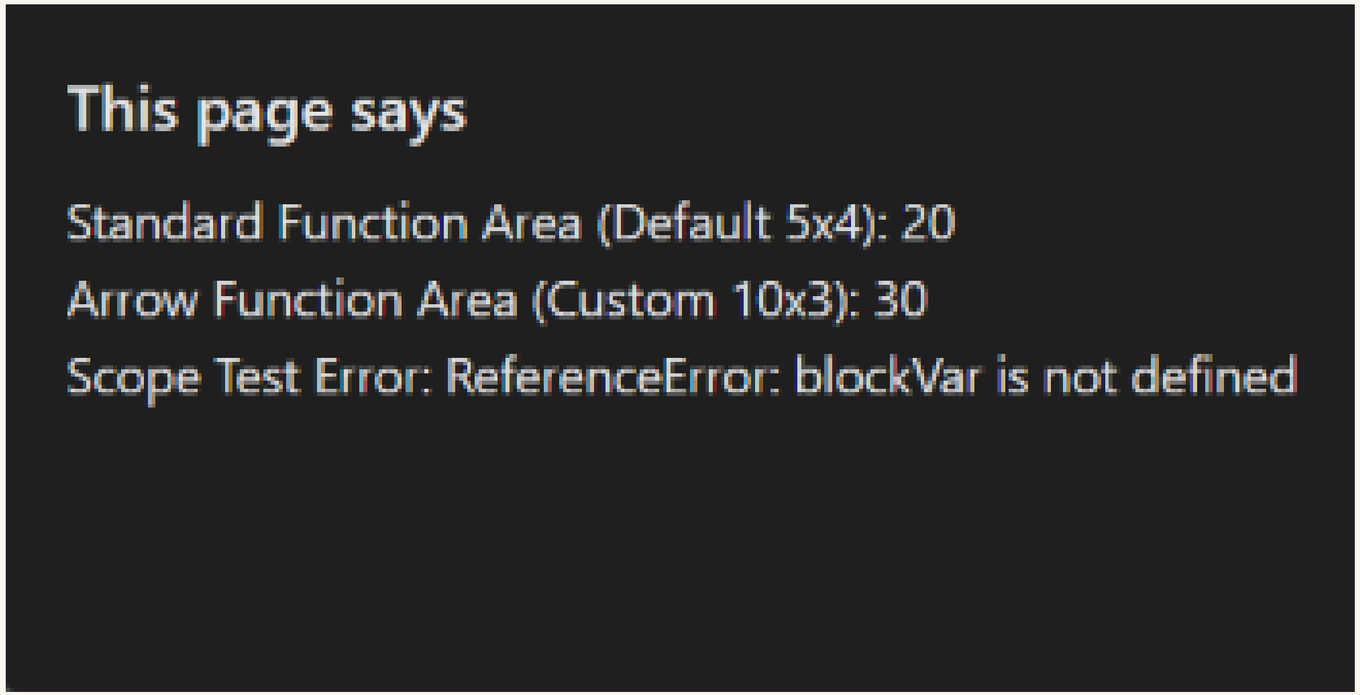


Assignment 5 : Function Architecture and Scope

Code

```
function calculateArea(length = 5, width = 4) {
    return length * width;
}
const calculateAreaArrow = (length = 5, width = 4) => {
    return length * width;
};
console.log(
    "Standard Function Area (Default 5x4): " +
    calculateArea()
);
console.log(
    "Arrow Function Area (Custom 10x3): " +
    calculateAreaArrow(10, 3)
);
{
    let blockVar = 100;
    console.log("Inside Block: " + blockVar);
}
let scopeMessage;
try {
    console.log(blockVar);
} catch (error) {
    scopeMessage = "Scope Test Error: ReferenceError: blockVar is not defined";
    console.log(scopeMessage);
}
alert(
    "Standard Function Area (Default 5x4): " + calculateArea() +
    "\nArro let scopeMessage: string | undefined ateAreaArrow(10, 3) +
    "\n" + scopeMessage
);
```

output



Assignment 6 : Arrays,Objects,and Core Methods

Code

```
const products = [
    { name: "Laptop", price: 800 },
    { name: "Mouse", price: 25 },
    { name: "Keyboard", price: 45 }
];
let output = "-- All Products (forEach) --\n";
products.forEach(function(product) {
    output += "Item: " + product.name +
              " - Price: $" + product.price + "\n";
});
output += "\n-- Filtered Products (Price < $50) --\n";
const filteredProducts = products.filter(function(product) {
    return product.price < 50;
});
output += JSON.stringify(filteredProducts, null, 2);
output += "\n\n-- Mapped Product Tags (map) --\n";
const mappedProducts = products.map(function(product) {
    return product.name.toUpperCase() +
           " costs $" + product.price;
});
output += JSON.stringify(mappedProducts, null, 2);
document.body.innerHTML =
    "<pre style='font-size:20px; white-space:pre-wrap; padding:20px;'>" +
    output + "</pre>";
console.log(output);
```

output

```
-- All Products (forEach) --
Item: Laptop - Price: $800
Item: Mouse - Price: $25
Item: Keyboard - Price: $45

-- Filtered Products (Price < $50) -
[
  {
    "name": "Mouse",
    "price": 25
  },
  {
    "name": "Keyboard",
    "price": 45
  }
]

-- Mapped Product Tags (map) --
[
  "LAPTOP costs $800",
  "MOUSE costs $25",
  "KEYBOARD costs $45"
]
```

Capstone Mini project : Arrays,Objects,and Core Methods

Code

```
const inventory = [
    { name: "Laptop", price: 800 },
    { name: "Mouse", price: 25 }
];
let productName = prompt("Enter new product name:", "Monitor");
let productPrice = Number(prompt("Enter product price:", "150"));
if (
    productName === null ||
    productName.trim() === "" ||
    !Number.isFinite(productPrice) ||
    productPrice <= 0
) {
    alert("Invalid product details. Please enter a valid name and price.
} else {
    inventory.push({
        name: productName.trim(),
        price: productPrice
    });
    function getCategory(price) {
        if (price < 50) {
            return "Budget";
        } else if (price < 200) {
            return "Standard";
        } else {
            return "Premium";
        }
    }
    const calculateTotal = (items) => {
        let total = 0;
        for (let item of items) {
            if (item.price <= 0) {
                continue;
            }
            total += item.price;
        }
        return total;
    };
    function applyDiscount(amount, discount = 0) {
        return amount - (amount * discount / 100);
    }
    let output = "--- Initializing Inventory System ---\n";
    output += "New product added: " + productName.trim() + "\n\n";
    output += "--- Processing Inventory Roster (forEach) ---\n";
    inventory.forEach(function(item) {
        output += "* Item: " + item.name +
```

```
    }
    let output = "--- Initializing Inventory System ---\n";
    output += "New product added: " + productName.trim() + "\n\n";
    output += "--- Processing Inventory Roster (forEach) ---\n";
    inventory.forEach(function(item) {
        output += "* Item: " + item.name +
                  " ($" + item.price + ") -> Category: " +
                  getCategory(item.price) + "\n";
    });
    const total = calculateTotal(inventory);
    const affordableItems = inventory.filter(function(item) {
        return item.price < 200;
    });
    const report = inventory.map(function(item) {
        return item.name.toUpperCase() + ": $" + item.price;
    });
    output += "\n--- Financial Analytics ---\n";
    output += "Total Inventory Value: $" + total + "\n";
    output += "After 10% Discount: $" +
              applyDiscount(total, 10) + "\n";
    output += "Filtered Affordable Items (Under $200): " +
              JSON.stringify(affordableItems.map(item => item.name)) + "\n";
    output += "Formatted Report List: " +
              JSON.stringify(report);
    console.log(output);
    document.body.innerHTML = "<pre style='font-size:18px; white-space:pre-wrap; padding:20px
    '>" + output + "</pre>";
console.log(output);
}
```

output

```
--- Initializing Inventory System ---
New product added: Monitor

--- Processing Inventory Roster (forEach) ---
* Item: Laptop ($800) -> Category: Premium
* Item: Mouse ($25) -> Category: Budget
* Item: Monitor ($150) -> Category: Standard

--- Financial Analytics ---
Total Inventory Value: $975
After 10% Discount: $877.5
Filtered Affordable Items (Under $200): ["Mouse","Monitor"]
Formatted Report List: ["LAPTOP: $800","MOUSE: $25","MONITOR: $150"]
```